MODULE -2 STRINGS

Python

Python Strings

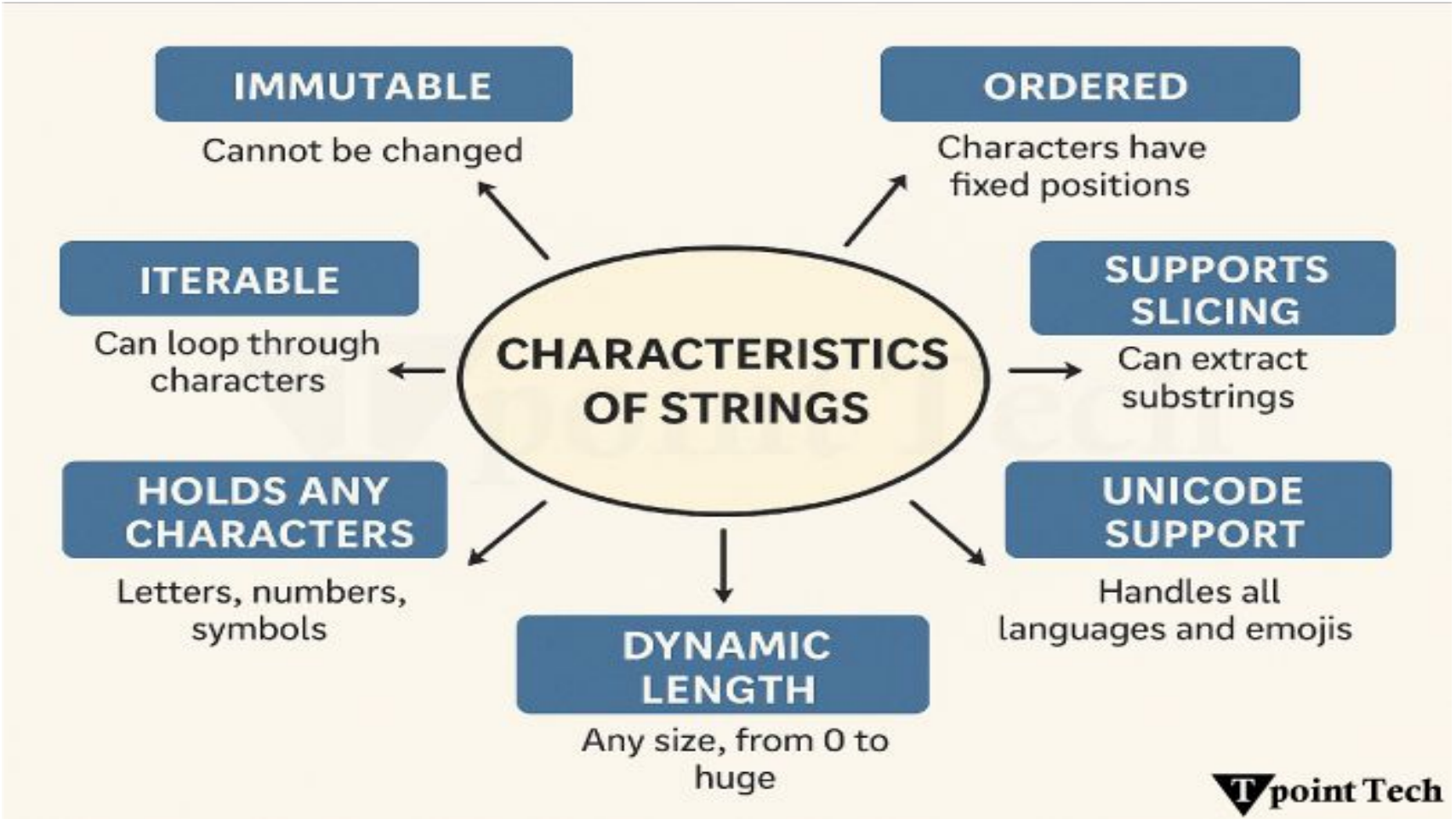
- A **String** in **Python** is a sequence of characters.
- includes letters, numbers, symbols, and even whitespaces.
- Python *does not have a* **character data type**; therefore, a **single character is considered as a string of length 1**.
- Syntax:

  Variable_name = “data” / Variable_name = ‘data’

Example:

```
greet = “Hello World”
string = ‘Welcome’
//multi line string
```

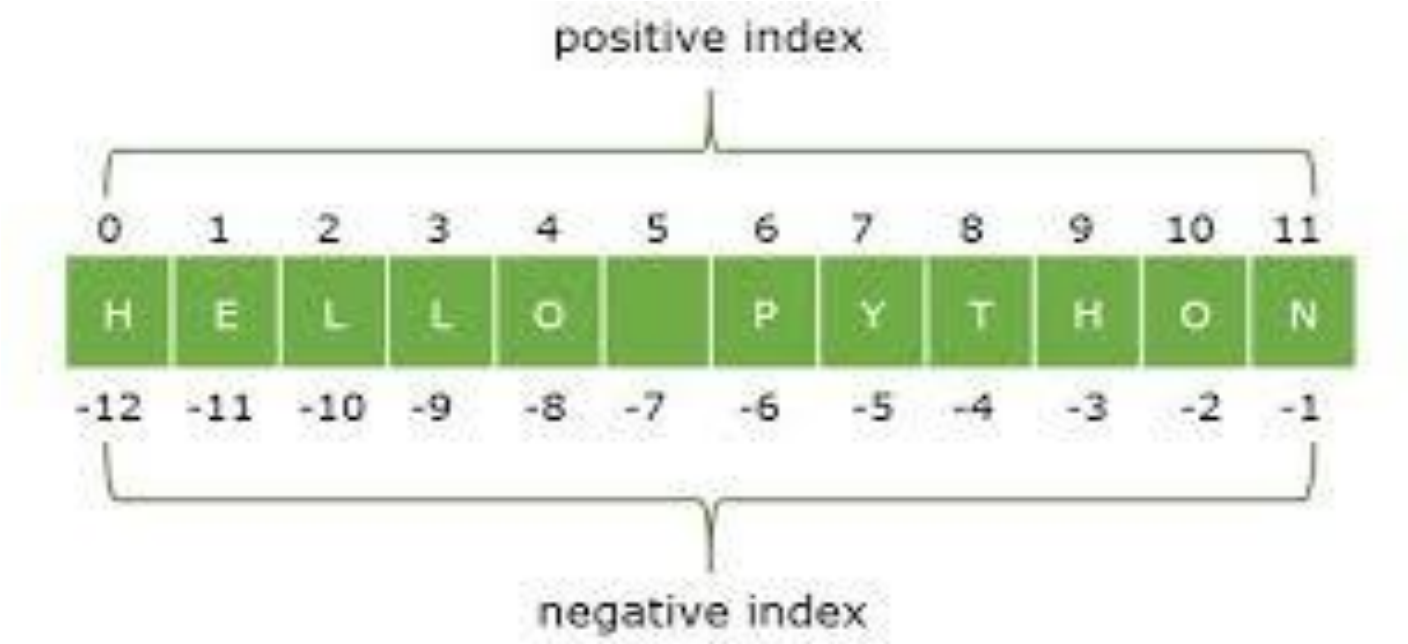
Characteristics of Strings



INDEXING

- Strings can be accessed individually with the help of indexing.
- Strings are indexed **0 from the start** and **-1 from the end**.

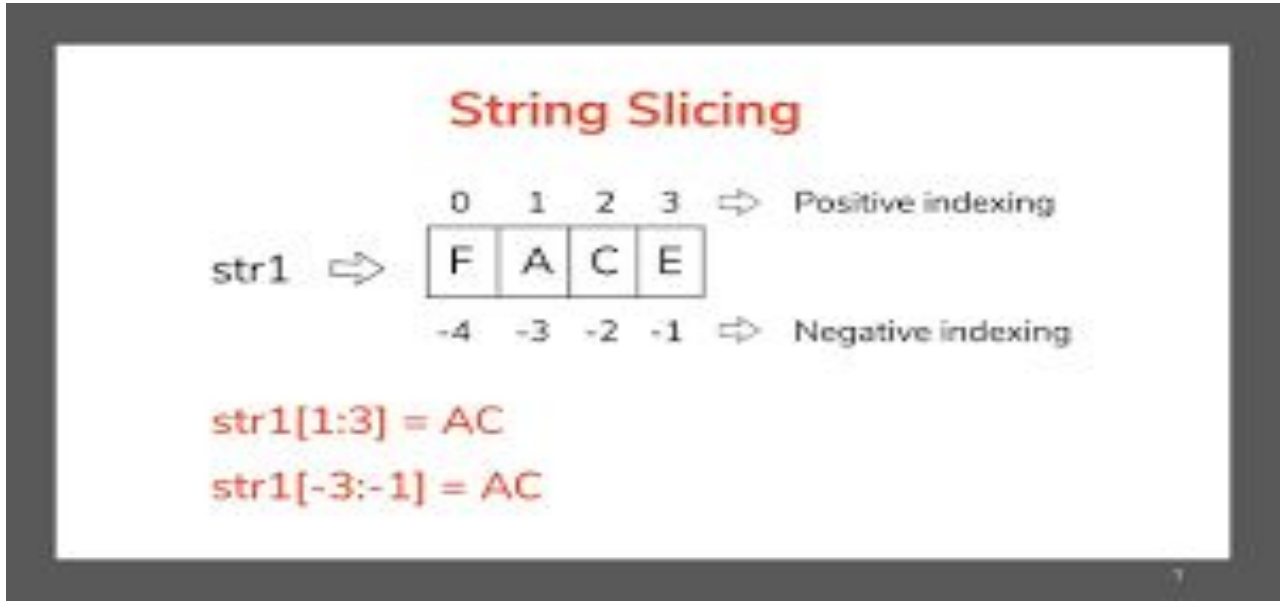
Example: Greet = “HELLO PYTHON”



STRING SLICING

- **Slicing** is a way in Python that allows us to extract a portion of a string by specifying the **start** and **end** indexes.
- Syntax:

  String_name[start:end:step]



String Methods in Python

- Built-in methods for string manipulation(like converting case, searching, splitting, joining, or checking)
- There are mainly 8 categories of string methods.

1. Case Conversion Methods
2. Searching and Finding
3. Validation / Checking
4. Modification / Replacement
5. Formatting and Alignment
6. Splitting and Joining
7. Encoding and Decoding
8. Miscellaneous

String Methods in Python

1. Case Conversion Methods

Method	Description	Example
upper()	Converts all characters to uppercase	"hello".upper() → 'HELLO'
lower()	Converts all to lowercase	"HELLO".lower() → 'hello'
title()	Converts first letter of each word to uppercase	"python programming".title() → 'Python Programming'
capitalize()	Converts first letter of string to uppercase	"hello world".capitalize() → 'Hello world'
swapcase()	Swaps uppercase ↔ lowercase	"PyThOn".swapcase() → 'pYtHoN'

String Methods in Python

2. Searching and Finding

Method	Description	Example
find(sub)	Returns index of first occurrence (or -1 if not found)	"hello".find('l') → 2
rfind(sub)	Finds last occurrence	"hello".rfind('l') → 3
index(sub)	Like find() but raises error if not found	"hello".index('l') → 2
rindex(sub)	Like rfind() but raises error	"hello".rindex('l') → 3
count(sub)	Counts occurrences of substring	"banana".count('a') → 3

String Methods in Python

3. Validation / Checking

Method	Description	Example
isalpha()	Only letters	"abc".isalpha() → True
isdigit()	Only digits	"123".isdigit() → True
isalnum()	Letters or digits	"abc123".isalnum() → True
isspace()	Only whitespace	" ".isspace() → True
isupper()	All uppercase	"HELLO".isupper() → True
islower()	All lowercase	"hello".islower() → True
istitle()	Title case	"Hello World".istitle() → True
startswith(sub)	Starts with substring	"python".startswith("py") → True
endswith(sub)	Ends with substring	"python".endswith("on") → True

String Methods in Python

4. Modification / Replacement

Method	Description	Example
replace(old, new)	Replace substring	"hello world".replace("world","Python") → 'hello Python'
strip()	Remove whitespace from both ends	" hello ".strip() → 'hello'
lstrip()	Remove leading spaces	" hello".lstrip() → 'hello'
rstrip()	Remove trailing spaces	"hello ".rstrip() → 'hello'
removeprefix("py")	Removes prefix (Python 3.9+)	"python".removeprefix("py") → 'thon'
removesuffix("on")	Removes suffix (Python 3.9+)	"python".removesuffix("on") → 'pyth'

String Methods in Python

5. Formatting and Alignment

Method	Description	Example
center(width, fillchar)	Centers text	"hi".center(10, '*') → '****hi****'
ljust(width, fillchar)	Left-align	"hi".ljust(10, '-') → 'hi--------'
rjust(width, fillchar)	Right-align	"hi".rjust(10, '-') → '--------hi'
zfill(width)	Pad with zeros	"42".zfill(5) → '00042'
format()	Format placeholders	"My name is {}".format("Raj") → 'My name is Raj'
f-string	Modern string formatting	f"My name is {name}"

String Methods in Python

6. Splitting and Joining

Method	Description	Example
split()	Splits into list by spaces	"a b c".split() → ['a','b','c']
split(',')	Splits by comma	"a,b,c".split(',') → ['a','b','c']
rsplit()	Split from right	"a,b,c".rsplit(',',1) → ['a,b','c']
partition(sep)	Splits into 3 parts (before, sep, after)	"name=raj".partition('=') → ('name', '=', 'raj')
rpartition(sep)	Same, but from right	"name=raj=123".rpartition('=') → ('name=raj', '=', '123')
join(iterable)	Joins list into string	" ".join(['a','b','c']) → 'a b c'

String Methods in Python

7. Encoding and Decoding

Method	Description	Example
encode()	Encode to bytes	"hello".encode() → b'hello'
decode()	Decode bytes to string	b'hello'.decode() → 'hello'

String Methods in Python

8. Miscellaneous

Method	Description	Example
expandtabs(n)	Replace \t with spaces	"a\tb".expandtabs(4) → 'a b'
translate(map)	Translate characters	"abc".translate({97: 'x'}) → 'xbc'
maketrans()	Create translation table	str.maketrans("abc","xyz")

Escape Sequences

- Print(“ Hello world”)
- Print(“He said “Hello””)
- **Syntax Error** as it contains both single and double-quotes.
- The backslash(/) symbol denotes the escape sequence.
- The backslash can be followed by a special character and it interpreted differently.

Escape Sequence	Meaning	Example Output
\n	New line	Moves text to the next line
\t	Tab space	Adds a horizontal tab
****	Backslash	Prints a single backslash \
\'	Single quote	Prints ' inside single-quoted strings
\"	Double quote	Prints " inside double-quoted strings
\r	Carriage return	Moves cursor to the beginning of the line
\b	Backspace	Removes one character backward